YEAR 10 PHYSICS, AQA GCSE, LESSON 10 OF 12

ELECTRIC CIRCUITS

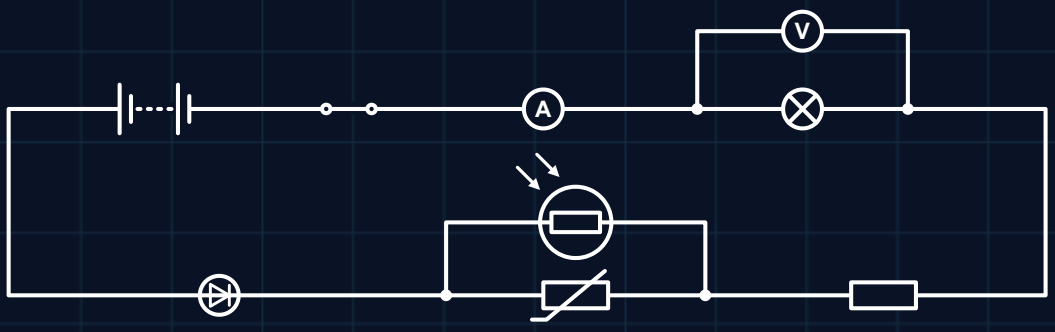
LESSON 10 THERMISTORS, LDRS AND SENSORS

How does a streetlight know it's getting dark, and how does a fire alarm know the room is hot?

IN THIS LESSON YOU WILL

- describe how a thermistor's resistance changes with temperature
- describe how an LDR's resistance changes with light intensity
- predict the current in a sensor circuit
- choose the right sensor for a job

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 10** of the Electric Circuits unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try “Your turn” alone.

TRY IT
Practice that builds up: first with support, then on your own. Quick checks from the slides follow some lessons.

LAB
Record measurements from a virtual lab or a required practical.

EXAM CORNER
An AQA-style question with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the “Cover the page” questions in Revise it. (3) Check with the quick answers at the back.

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Complete circuits and symbols

02 Current and charge

03 Potential difference

04 Resistance and V = I R

05 Resistance of a wire RP

06 Series and parallel circuits

07 Resistors in series and parallel RP

08 I–V graphs

09 Investigating I–V characteristics RP

10 Thermistors, LDRs and sensors
How thermistors and LDRs respond to temperature and light, and sensor circuits

11 Power and energy transfer

12 Mains, safety and the National Grid

Lesson

10 THERMISTORS, LDRS AND SENSORS

AQA Combined 6.2.1.4, 6.2.2; Physics 4.2.1.4, 4.2.2

BIG QUESTION **How does a streetlight know it's getting dark, and how does a fire alarm know the room is hot?**

BY THE END YOU CAN

- describe how a thermistor's resistance changes with temperature
- describe how an LDR's resistance changes with light intensity
- predict the current in a sensor circuit
- choose the right sensor for a job

thermistor **LDR** **light intensity** **sensor**

RETRIEVE

DO NOW

From earlier lessons. No notes!

1
Which component's I–V graph gets less steep?

Filament lamp.

2
At a fixed p.d., what happens to the current if the resistance doubles?

It halves.

3
Draw the symbol for a diode.

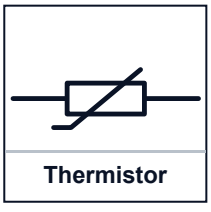
A triangle pointing at a bar, on a line.

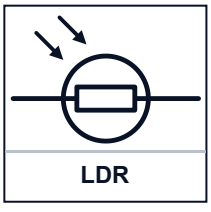
EXPLAIN

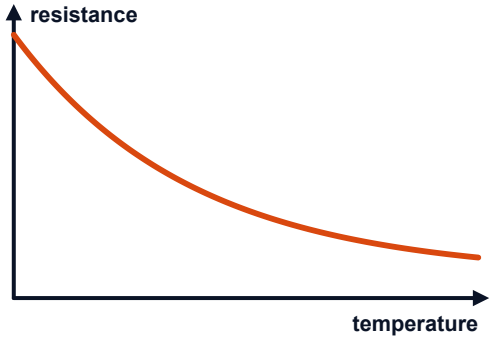
COMPONENTS THAT SENSE

Some resistors change their resistance when their surroundings change. That makes them useful as **sensors**.

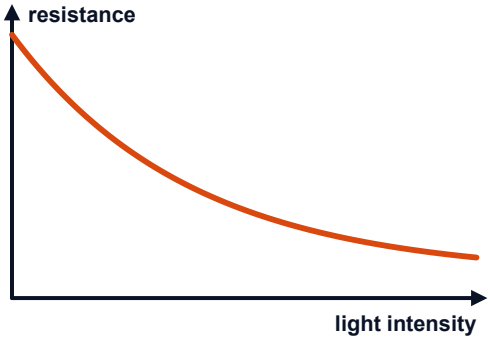
A **thermistor** responds to **temperature**. A **light-dependent resistor (LDR)** responds to **light intensity** (how bright it is).


Thermistor


LDR



Thermistor: as the temperature increases, its resistance decreases.



LDR: as the light gets brighter, its resistance decreases.

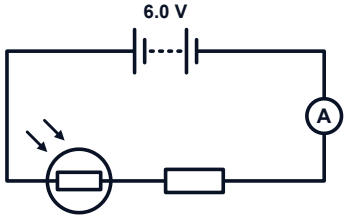
! Both follow **more → less**: more heat or more light means less resistance.

	THERMISTOR	LDR
Responds to	temperature	light intensity
When that increases…	resistance decreases	resistance decreases
Used in	thermostats, fire alarms, car engine sensors	night lights, streetlights, security lights

SENSORS IN CIRCUITS

In a series circuit with a fixed supply p.d., the current depends on the **total resistance**. When light shines on the LDR, its resistance falls, so the total resistance falls and the **current rises** (I = V ÷ R).

A change in current (or in the p.d. across the fixed resistor) can be used to switch something on, such as a lamp or an alarm.



An LDR in series with a fixed resistor.

WORKED EXAMPLE

A 6.0 V supply is connected to an LDR and a resistor in series. In bright light the total resistance is 600 Ω. Calculate the current.

EQUATION I = V ÷ R

SUBSTITUTE I = 6.0 ÷ 600

ANSWER **I = 0.010 A (10 mA)**

YOUR TURN

In the dark, the total resistance rises to 1500 Ω. Calculate the new current.

EQUATION **I = V ÷ R**

SUBSTITUTE **I = 6.0 ÷ 1500**

ANSWER **I = 0.0040 A (4.0 mA)**

PRACTISE

TRY IT

1 Complete the sentences.

When the temperature of a thermistor increases, its resistance **decreases**. When the light on an LDR gets brighter, its resistance **decreases**. So in the dark, an LDR has a **high** resistance.

WORD BANK: **increases** | **decreases** | **high** | **low**

2 Choose a thermistor or an LDR for each job, and give a reason.

Job	Sensor	Why
Fire alarm	**thermistor**	**detects a rise in temperature**
Automatic garden light	**LDR**	**detects when it gets dark**
Greenhouse heater that switches on when it is cold	**thermistor**	**detects a fall in temperature**
Camera that measures how bright a scene is	**LDR**	**detects light intensity**

3 A thermistor is in series with an ammeter and a 12 V supply. The thermistor is put into hot water. Describe and explain what happens to the ammeter reading.

The reading increases. The thermistor gets hotter, so its resistance decreases, the total resistance decreases, and I = V ÷ R increases.

4 SUPPORTED A 12 V sensor circuit has a total resistance of 300 Ω. Calculate the current.

EQUATION	**I = V ÷ R**
SUBSTITUTE	**I = 12 ÷ 300**
ANSWER	**0.040 A** unit:

5 Which change would **increase** the current through a thermistor in a series circuit?

- ☐ cooling the thermistor
- ☐ covering it with a black cloth
- ■ heating the thermistor
- ☐ adding another resistor in series

CHECKPOINT

QUICK CHECK

Slide 40, Mission 6. Answer each question, then check it as your teacher reveals the answer.

1
What does a straight-line I–V graph through the origin show?

An ohmic conductor, such as a fixed resistor at constant temperature.

2
Why does the I–V graph of a lamp curve?

The filament heats up, so its resistance increases.

3
What happens with a diode in reverse?

Its resistance is very high, so there is almost no current.

4
A thermistor gets hotter. What happens to its resistance?

It decreases.

5
Which sensor would switch on a streetlight?

An LDR.

6
The supply is reversed. What happens to the readings?

The current flows the other way, so the readings are negative.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** For ‘explain’, link each step: light → LDR resistance → total resistance → current.

6 A 6.0 V supply is connected to an LDR and a fixed resistor in series. In dim light the total resistance is 1500 Ω. (a) Calculate the current. (b) Explain what happens to the current when the light becomes brighter. **[4 marks]**

(a) I = V ÷ R (1) = 6.0 ÷ 1500 = 0.0040 A (1). (b) The LDR's resistance decreases (1), so the total resistance decreases and the current increases (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Thermistor: temperature up → resistance **down**.
- LDR: light intensity up → resistance **down**.
- In a series circuit at a fixed p.d.: lower total resistance → bigger current.
- Thermistors sense temperature (thermostats, fire alarms); LDRs sense light (streetlights, night lights).

COVER THE PAGE AND ANSWER

1 What happens to a thermistor's resistance as it warms up?

2 What happens to an LDR's resistance in the dark?

3 Which sensor would you use in a streetlight?

4 Total R falls in a series circuit. What happens to the current?

I CAN…

- describe how a thermistor responds to temperature
- describe how an LDR responds to light
- explain how the current in a sensor circuit changes
- choose and justify a sensor for a job

REFERENCE

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: AQA gives you an equation sheet in the exam (it does not include $R_{total} = R_1 + R_2$), but knowing the equations by heart makes every question faster.

$Q = I\,t$

Lesson 2

charge flow = current × time

C, A, s

Rearranged: I = Q ÷ t and t = Q ÷ I

$E = Q\,V$

Lesson 3

energy transferred = charge × p.d.

J, C, V

Rearranged: V = E ÷ Q and Q = E ÷ V

$V = I\,R$

Lesson 4

p.d. = current × resistance

V, A, Ω

Rearranged: I = V ÷ R and R = V ÷ I

$R_{total} = R_1 + R_2$

Lesson 6

total resistance in series

Ω

Rearranged: (series only)

FOUR STEPS FOR EVERY CALCULATION

1
Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2
Write the equation you will use, in symbols.

3
Substitute the numbers, then rearrange if needed.

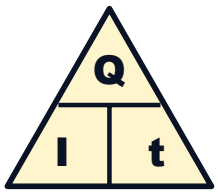
4
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes → seconds	× 60	3.0 min = 180 s
hours → seconds	× 3600	2.0 h = 7200 s
milliamps (mA) → amps (A)	÷ 1000	250 mA = 0.25 A
kilowatts (kW) → watts (W); kilohms (kΩ) → ohms (Ω)	× 1000	2.2 kW = 2200 W
kilovolts (kV) → volts (V)	× 1000	400 kV = 400 000 V

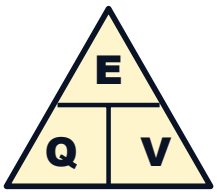
EQUATION TRIANGLES: HELP WITH REARRANGING

1. Cover the quantity you want to find.
2. The two left **side by side**? **Multiply** them.
3. One **above** the other? **Divide** the top by the bottom.

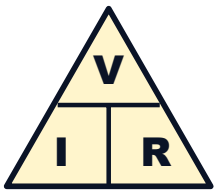
Example: to find I from Q = I t, cover I. Q is above t, so I = Q ÷ t. Triangles only help with rearranging: you still need to learn each equation.



Q = I t
Q = I × t
I = Q ÷ t
t = Q ÷ I



E = Q V
E = Q × V
Q = E ÷ V
V = E ÷ Q



V = I R
V = I × R
I = V ÷ R
R = V ÷ I

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

LDR: Light-dependent resistor: resistance decreases as light intensity increases.

Thermistor: Resistance decreases as temperature increases.

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

DO NOW

1. Filament lamp.
2. It halves.
3. A triangle pointing at a bar, on a line.

COVER THE PAGE

1. It decreases.
2. It increases (it is high).
3. An LDR.
4. It increases.